# CSCI 112 Assignment – Arrays and Loops

Points: 10

Automated tests will run on GitHub Actions after every push. Check the Actions tab to see which tests pass and which fail.

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## Instructions

1. **Fork** the repository on GitHub - `https://github.com/jpach-cs/112-ass05-template`.
2. Clone your forked repository to your local machine.
3. Open the file `code.h` and add:
    1. Include guards at the top: `#ifndef CODE_H` `#define CODE_H`
    2. All function prototypes
    3. Include guards at the bottom: `#endif`
4. Open the file `code.c`.
    1. Implement every function listed below:
        - Read the comments above each function carefully.
        - Follow the rules — use ONLY `if`, `switch`, `while`, `for`, `putchar()`, `getchar()`, `printf()`, `sprintf()`.
        - Do NOT use `goto`, or any standard library functions.
    2. Test your implementation locally using `main.c`.
5. Once all functions are complete — save your changes, then **commit** and **push** to your repository.
6. Go to the **Actions** tab on GitHub and check the test results.

### Learning Objectives

After completing this lab, you will:

Use `for` loops to iterate over arrays with known length Use `while` loops to search and converge on solutions Implement array manipulation functions (`min`, `max`, `sum`, `average`) Implement search algorithms (`linear_search`) Implement iterative mathematical algorithms (`heron` - Heron's method) Work with multi-file C projects using header files with include guards.

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## Implement 6 functions that work with arrays and use iteration:

| Function | Purpose | Loop Type | Return Type |
|---|---|---|---|
| `find_max()` | Find largest element | `for` | `int` |
| `find_min()` | Find smallest element | `for` | `int` |
| `sum_array()` | Sum all elements | `for` | `long` |

| Function | Purpose | Loop Type | Return Type |
|---|---|---|---|
| `average()` | Calculate mean | `for` | `double` |
| `linear_search()` | Find element index | `while` | `int` |
| `heron()` | Calculate square root | `while` | `double` |

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# Background

## For Loops vs While Loops

Use `for` loops when you know the iteration count:

```
for (int i = 0; i < n; i++)
{
    // Process array[i]
}
```

Use `while` loops when you don't know when to stop:

```
int i = 0;
while (i < n)
{
    if (/* found it */)
        return i;
    i++;
}
```

## Include Guards

Header files must use include guards to prevent multiple inclusion:

```
#ifndef CODE_H
#define CODE_H

int find_max(int arr[], int n);
// ... more prototypes

#endif
```

## Heron's Method for Square Root

Background:

Heron's method iteratively improves an initial guess:

```
next_guess = (current_guess + x / current_guess) / 2.0
```

Stop when consecutive guesses differ by less than epsilon.

Algorithm:

1. Start with initial guess: guess = x / 2.0
2. Use while (1) loop (infinite loop with break):
   - Save current guess: prev_guess = guess
   - Calculate next guess: guess = (guess + x / guess) / 2.0
   - Calculate difference: diff = guess - prev_guess
   - Make difference positive manually: if (diff < 0) diff = -diff;
   - If diff < epsilon, break the loop
3. Return the final guess

Note: You cannot use fabs() - manually check if negative with if (x < 0) x = -x;

Example: heron(4.0, 0.0001); // Returns 2.0 heron(9.0, 0.0001); // Returns 3.0 heron(2.0, 0.0001); // Returns 1.414

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### #Header comment — required in code.c

```
// Your Name
// CSCI 112 Fall 2026
// Programming Assignment #1
// I declare that I am the author of this work, take full responsibility for it, and have d
```

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## Only for testing

### Local Testing (Sandbox)

Create test cases in main.c:

```c
#include <stdio.h>
#include "code.h"

int main()
{
    int arr[] = {5, 2, 8, 1, 9};
    printf("Max: %d\n", find_max(arr, 5));
    printf("Min: %d\n", find_min(arr, 5));
    printf("Sum: %ld\n", sum_array(arr, 5));

    float farr[] = {2.0f, 4.0f, 6.0f};
    printf("Average: %f\n", average(farr, 3));
```

```
    printf("Search for 8: %d\n", linear_search(arr, 5, 8));
    printf("Search for 7: %d\n", linear_search(arr, 5, 7));

    printf("Square root of 4: %f\n", heron(4.0, 0.0001));
    printf("Square root of 2: %f\n", heron(2.0, 0.0001));

    return 0;
}
```